

Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing

Objectives

1. To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.
2. To compare the slot duration corresponding to different Subcarrier Spacing values in 5G NR.
3. To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

Objective – 1: Analyze Different Subcarrier Spacing Values MATLAB Program

MATLAB Program

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

mu = 0:4; % Numerology Index

bar(scs)

grid on

set(gca,'XTick',1:5)

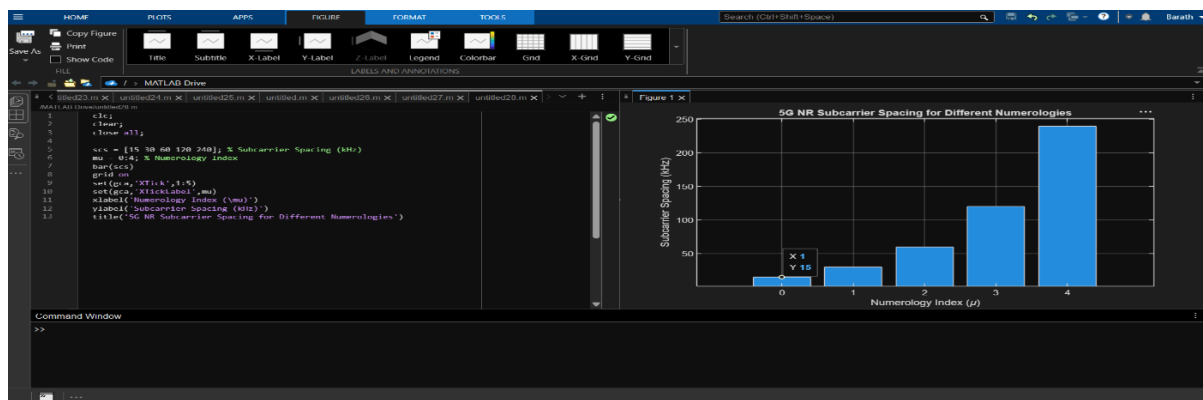
set(gca,'XTickLabel',mu)

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

title('5G NR Subcarrier Spacing for Different Numerologies')
```

OUTPUT

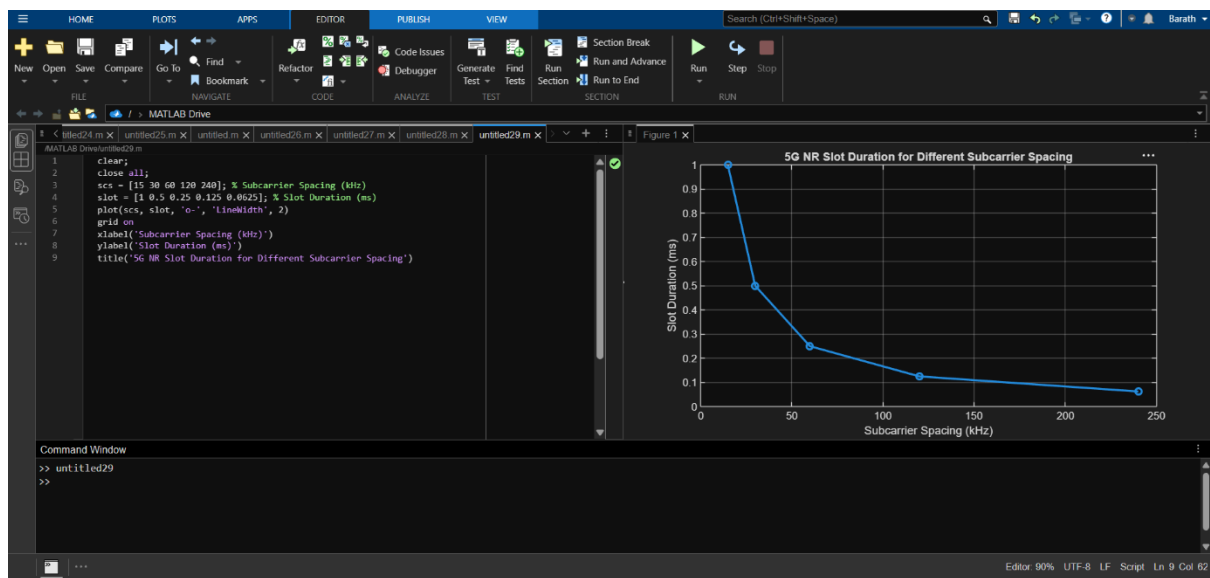


Objective – 2 Compare Slot Duration

MATLAB Program

```
clear;close all;  
  
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)  
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)  
  
plot(scs, slot, 'o-', 'LineWidth', 2)  
  
grid on  
  
xlabel('Subcarrier Spacing (kHz)')  
  
ylabel('Slot Duration (ms)')  
  
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

OUTPUT



Objective – 3 Study the Number of OFDM Symbols

MATLAB Program

```
clc;  
  
clear;  
  
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
```

```
figure
```

```
bar(scs, symbols)
```

```
grid on
```

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Number of OFDM Symbols')
```

```
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
```

```
ylim([0 16])
```

OUTPUT

